GeulPU 0.21.0 - Independent Correctness Test Summary

5-6 November 2026 - engine commit 0efe2a5824ea202f5c02f98afb0620647966e884 - independent test results; not measurements reported by the engine authors

This is an independent validation run by researchers from GING (Next Generation Internet Group), Universidad Politécnica de Madrid (UPM): https://ging.github.io

Test (all token/logit comparisons)	RTX 6000 Ada full GPU, 48 GB	H100 NVL MIG 3g.47gb, 47 GB	A100 MIG 7g.40gb, 40 GB
1. Official repro kit: 144 requests identical to kit references (Qwen3-8B Q4_K_M and Q8_0, Qwen3.5-9B Q4_K_M)	**PASS** 144/144	**PASS** 144/144	**PASS** 144/144
1. Official kit built-in CUDA check (DeepSeek-R1-Distill-Qwen-7B)	**PASS** 32/32	**PASS** 32/32	**PASS** 32/32
2. Core, Qwen3-8B Q4_K_M: outputs vs own cold references	**PASS** 36/36	**PASS** 36/36	**PASS** 36/36
2. Core: request overlap	**PASS** 15/15	**PASS** 15/15	**PASS** 15/15
2. Core: 16-request 4-way burst	**PASS**	**PASS**	**PASS**
2. Core follow-up: cache eviction + SSD restore	**PASS** 4/4	**PASS** 4/4	**PASS** 4/4
2. Cross-GPU agreement of core outputs (one multi-GPU observation, not 3 independent passes)	**PASS** 58/58 matched across the three allocations (observed agreement)		
3. Qwen3.5-9B Q4_K_M (hybrid): outputs vs cold references	**PASS** 36/36	**PASS** 36/36	**PASS** 36/36
3. Qwen3-8B Q8_0: outputs vs cold references	**PASS** 37/37	**PASS** 37/37	**PASS** 37/37
3. Both: tokenization parity 17/17, overlap 15/15, 17-request 4-way	**PASS**	**PASS**	**PASS**
3. Qwen3-8B Q8_0: SSD restore after eviction	**PASS**	**PASS**	**PASS**
4. Spec ON vs OFF, Qwen3-8B Q8_0: within-arm 37/37 each	**PASS** 37/37 ON, 37/37 OFF	**PASS** 37/37 ON, 37/37 OFF	**PASS** 37/37 ON, 37/37 OFF
4. Spec ON vs OFF: cross-arm identity (60 compared rows, reused requests, not 60 independent repeats)	**PASS** 60/60	**PASS** 60/60	**PASS** 60/60
5. Eight-way concurrency, Qwen3-8B Q8_0: 23 requests, slots 0-7, context 8,192	**PASS** 23/23	**PASS** 23/23	**PASS** 23/23
6. Second family, Llama-3.1-8B-Instruct Q8_0, speculation OFF suite: 37/37 vs own cold references; overlap 15/15, 4-way, SSD restore	**PASS** 37/37	**PASS** 37/37	**PASS** 37/37
6. Llama-3.1-8B-Instruct Q8_0: eight-way concurrency, 23 requests, slots 0-7, context 8,192	**PASS** 23/23	**PASS** 23/23	**PASS** 23/23
6. Llama-3.1-8B-Instruct Q8_0: speculation ON comparison (within-arm 37/37; cross-arm 60 rows vs OFF)	**PASS** 37/37 ON, 60/60 cross	**PASS** 37/37 ON, 60/60 cross	**PASS** 37/37 ON, 60/60 cross
6. Cross-GPU agreement of Llama speculation-OFF outputs (one multi-GPU observation, not 3 independent passes; distinct from the Qwen 58/58 core observation above)	**PASS** 23/23 matched across the three allocations (observed agreement)		
KV8 / fast attention	Engaged only in core Q4; N/A (unsupported) on Q8, Qwen3.5, Llama	Engaged only in core Q4; N/A (unsupported) on Q8, Qwen3.5, Llama	Engaged only in core Q4; N/A (unsupported) on Q8, Qwen3.5, Llama
Speculation / SSD store on Qwen3.5 (hybrid)	N/A (engine capability)	N/A (engine capability)	N/A (engine capability)

Legend: green = PASS (completed run, all checks identical); red = FAIL (none occurred in the completed suites shown); gray = not run on that allocation; light = N/A by feature or engine capability. MIG slices are partitioned instances, not whole-GPU rankings.

Models tested (GGUF files; model - quantization - where used)

- Qwen3-8B, Q4_K_M: official kit and core suite (section 2)
- Qwen3-8B, Q8_0: official kit and additional suite (sections 3-5)
- Qwen3.5-9B (hybrid), Q4_K_M: official kit and additional suite (section 3)
- DeepSeek-R1-Distill-Qwen-7B, Q8_0: official kit built-in CUDA check only (section 1)
- Llama-3.1-8B-Instruct, Q8_0 (non-Qwen family): speculation-OFF suite, eight-way concurrency and speculation-ON comparison on all three allocations (section 6). Gemma 4 was not run: it is exact-mode only in this engine, outside the fast-mode scope

Test details

Official kit: 12 fixed requests, serial then 8 concurrent, compared on prompt token IDs, generated token IDs and per-step logit hashes against the author's RTX 4070 Ti references. Core follow-up: a separately reviewed run proved eviction and SSD restore (7,663 tokens restored after only 17 memory-cache tokens); the original core full runs remain INCOMPLETE for that path (cache pressure did not evict) and were not rewritten. Additional runs each passed an independent audit. Spec ON/OFF: OFF verified on all 5 model loads; the 60 rows are 23 cold + 16 churn + 4 eviction + 17 four-way reuses. Eight-way: 23/23 server completions, all 8 slots used, observed concurrency 8, no silent context shrink; sampled CUDA free-memory delta about 17.8 GiB (H100) and 18.0 GiB (A100) for Qwen, 16.6 GiB for Llama on H100 (RTX figure as reported earlier, ~18.5 GiB). Llama: a first H100 attempt stopped at the churn step (10/15 overlap vs required 14/15, because its resident answer ended after 10.9 s); it was not retried silently. A fresh run with 0.5 s request spacing and the 14/15 threshold unchanged then passed 15/15. The Llama speculation-OFF suite, eight-way concurrency and speculation-ON comparison then ran on all three allocations; every arm passed its independent audit (within-arm 37/37 ON and OFF, cross-arm 60/60, eight-way 23/23). The Llama eight-way on RTX 6000 Ada ran beside a live unrelated demo process: its pre-run memory gate passed with a 43 MiB margin and the sampled guard minimum free memory was 2,047 MiB (47 MiB above its trip level) - a legitimate PASS with tight margins, not an operational-margin recommendation; the demo stayed live throughout. Speculation-ON arms are reported as configured/initialized with effective flags verified three ways; their timings are descriptive only and no speed claim is made.

Scope and limits

- Scope is the completed correctness suites above, for the tested models, quantizations and allocations only. Gray cells were not tested; they are not failures.
- Custom suites (2-6) ran in default fast mode: observed within-run repeatability, empirical evidence, not a determinism guarantee. Exact mode and fault injection were excluded. The official kit used its own profiles as shipped.
- The Llama cross-GPU 23/23 row is one empirical observation (generated tokens and per-step logit vectors of 23 shared requests matched across H100, A100 and RTX), kept distinct from the Qwen 58/58 core observation; it is not a guarantee.
- KV8 and fast attention were requested for the additional-model suites but are unsupported on those paths, so inactive.
- Serialized tool-call text can differ between identical runs; where it did, prompt IDs, generated token IDs and logit hashes still matched. Claims are token/logit identity, not text identity.
- Timings are descriptive and unpaired; MIG memory telemetry was unavailable; memory figures are sampled CUDA free-memory deltas only. A sampled own-context memory guard was active only in the eight-way runs and the Llama full runs; no measured-guard claim is made for the other runs (including Qwen speculation-OFF arms and the Llama smokes).